Chapter 1: Gravitation

Q1. State Kepler's laws of planetary motion. Write any one law in words and explain its meaning in brief.

Kepler's laws of planetary motion are:

1) Law of Orbits: Each planet moves around the Sun in an elliptical orbit with the Sun at one focus.

2) Law of Areas: The line joining the planet and the Sun sweeps out equal areas in equal intervals of time.

3) Law of Periods: The square of the time period of revolution (T^2) is directly proportional to the cube of the mean distance from the Sun (r^3), i.e. $T^2 \propto r^3$.

Example (Law of Areas – meaning): When a planet is nearer to the Sun, it moves faster so that equal areas are swept in equal time. When it is farther, it moves slower.

Q2. State Newton's universal law of gravitation. Write its mathematical expression and explain the meaning of each symbol used.

Statement: Every two bodies in the universe attract each other with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.

Mathematical expression: $F = G\, m_1 m_2 / r^2$

Where,

F = gravitational force between the two bodies (in newton, N)

m_1, m_2 = masses of the bodies (in kilogram, kg)

r = distance between the centres of the two bodies (in metre, m)

G = universal gravitational constant ($G = 6.67 \times 10^{-11}$ N m²/kg²)

Q3. Using Newton's law of gravitation, derive the inverse square law of gravitation (relation between force and distance).

From Newton's law of gravitation: $F = G\, m_1 m_2 / r^2$

For fixed masses m_1 and m_2, and constant G, we have: $F \propto 1/r^2$

Thus, the gravitational force varies inversely as the square of the distance between the centres of the two bodies.

Verification in words: If distance becomes double ($r \rightarrow 2r$), then force becomes one-fourth ($F \rightarrow F/4$). If distance becomes three times, force becomes 1/9 times.

Q4. Derive the expression for centripetal force acting on a body moving in a circular path.

Consider a body of mass m moving with uniform speed v in a circular path of radius r.

Even though speed is constant, direction changes continuously, hence the body has centripetal acceleration towards the centre.

Centripetal acceleration: $a = v^2 / r$

Therefore, centripetal force: $F = m\,a = m\,(v^2 / r)$

So, Centripetal force, $F = m\,v^2 / r$

This force always acts towards the centre of the circular path.

Q5. Derive the expression for acceleration due to gravity (g) on the surface of the Earth.

Let a body of mass m be on the surface of Earth (mass of Earth = M, radius = R).

Gravitational force on the body due to Earth: $F = G\,M\,m / R^2$

Also, the weight of the body is: $F = mg$

Equating both forces: $mg = G\,M\,m / R^2$

Cancel m on both sides: $g = G\,M / R^2$

Thus, acceleration due to gravity on Earth's surface is: $g = GM/R^2$.

Q6. Explain why the value of g varies with altitude. Write the conclusion for g at higher altitude.

At a height h above Earth's surface, the distance from Earth's centre becomes (R + h).

Acceleration due to gravity at height h is: $g_h = GM / (R + h)^2$

Comparing with $g = GM/R^2$, we get: $g_h = g \times (R^2 / (R + h)^2)$

Since (R + h) > R, the denominator increases, so g_h decreases.

Conclusion: The value of g decreases as altitude increases.

Q7. Explain why the value of g varies with depth. Write the conclusion for g at the centre of the Earth.

As we go inside the Earth, the effective mass responsible for gravitational attraction decreases (only the mass within that radius acts).

At depth d below the surface, distance from centre is (R − d).

Acceleration due to gravity decreases with depth and becomes maximum at surface.
At the centre of the Earth, gravitational pull from all directions cancels out.
Conclusion: At the centre of Earth, g becomes zero.

Q8. If the mass of Earth becomes double and its radius becomes half, calculate the new value of g in terms of original g. Justify mathematically.
Given: $g = GM/R^2$
New mass: $M' = 2M$
New radius: $R' = R/2$
New g: $g' = G M' / (R')^2 = G (2M) / (R/2)^2$
$(R/2)^2 = R^2/4$
So, $g' = (2GM) / (R^2/4) = (2GM) \times (4/R^2) = 8GM/R^2$
Therefore, $g' = 8g$
New acceleration due to gravity becomes eight times the original value.

Q9. Differentiate between mass and weight on the basis of definition, SI unit and dependence on gravity.
Mass:
Definition: Amount of matter contained in a body.
SI unit: kilogram (kg).
Dependence on gravity: Mass is constant and does not change from place to place.
Weight:
Definition: Gravitational force with which Earth attracts the body ($W = mg$).
SI unit: newton (N).
Dependence on gravity: Weight depends on g, hence changes from place to place.

Q10. Explain why weight of an object changes from place to place but its mass remains constant.
Mass is the amount of matter in the object, so it remains the same everywhere.
Weight is given by $W = mg$, where g is acceleration due to gravity.
The value of g varies with altitude, depth, latitude and presence of other celestial bodies.
Therefore, weight changes from place to place because g changes, but mass remains constant.

Q11. Define free fall. State any two characteristics of a freely falling body.
Definition: Free fall is the motion of a body under the influence of gravity alone, with no other force (like air resistance) acting on it.
Characteristics (any two):
1) The body has constant acceleration equal to g directed towards the Earth's centre.
2) Its velocity increases uniformly with time while falling (in free fall).
3) All bodies fall with the same acceleration in free fall, irrespective of their masses (neglecting air resistance).

Q12. A stone is dropped from a height of 125 m. Calculate (a) time taken to reach the ground and (b) velocity with which it strikes the ground. ($g = 10\ m/s^2$)
Given: $u = 0$ (dropped), $s = 125$ m, $g = 10\ m/s^2$
(a) Time taken: Use $s = ut + (1/2)gt^2$
$125 = 0 + (1/2)(10)t^2 = 5t^2$
$t^2 = 25 \Rightarrow t = 5$ s
(b) Velocity on reaching ground: Use $v = u + gt$
$v = 0 + 10 \times 5 = 50$ m/s
Answer: Time = 5 s, Velocity = 50 m/s (downward).

Q13. A body is thrown vertically upward with initial velocity 20 m/s. Calculate (a) maximum height reached and (b) time taken to reach the highest point. ($g = 10\ m/s^2$)
Given: $u = 20$ m/s, $g = 10\ m/s^2$ (downward). At highest point, $v = 0$.
(a) Maximum height: Use $v^2 = u^2 - 2gh$
$0 = (20)^2 - 2(10)h$
$0 = 400 - 20h \Rightarrow h = 20$ m
(b) Time to reach highest point: Use $v = u - gt$
$0 = 20 - 10t \Rightarrow t = 2$ s
Answer: Maximum height = 20 m, Time = 2 s.

Q14. Two masses of 5 kg and 10 kg are kept 2 m apart. Calculate the gravitational force between them. ($G = 6.67 \times 10^{-11}\ N\ m^2/kg^2$)
Given: $m_1 = 5$ kg, $m_2 = 10$ kg, $r = 2$ m, $G = 6.67 \times 10^{-11}\ N\ m^2/kg^2$
Formula: $F = G\, m_1 m_2 / r^2$
$F = (6.67 \times 10^{-11}) \times (5 \times 10) / (2^2)$
$5 \times 10 = 50$ and $2^2 = 4$
$F = (6.67 \times 10^{-11}) \times (50/4) = (6.67 \times 10^{-11}) \times 12.5$
$F = 8.3375 \times 10^{-10}$ N
Answer: Gravitational force $\approx 8.34 \times 10^{-10}$ N.

Q15. Define escape velocity. Derive its expression and state its significance in space travel.
Definition: Escape velocity is the minimum velocity with which a body must be projected from the surface of a planet so that it can escape the planet's gravitational field and never return.
Derivation (outline):

To escape, the body must have enough kinetic energy to overcome gravitational potential energy.

At surface: K.E. = $(1/2)\, m\, v_e^2$

Gravitational potential energy needed to escape from radius R to infinity is: $U = GMm/R$

Equating for minimum escape: $(1/2)\, m\, v_e^2 = GMm/R$

Cancel m: $v_e^2 = 2GM/R \Rightarrow v_e = \sqrt{(2GM/R)}$

Significance:

Escape velocity decides the minimum launch speed required for rockets/satellites to leave Earth's gravitational pull.

It is important for space missions, launching probes and interplanetary travel.